

Technical Data Sheet

Date : 01.07.2025
Product Name : Transformer Oil
Grade : IEC 60296 & Customer Specification
Customer : NTPC

Rev. No.01

Sr. No.	Characteristics	Test Method	Specified Values	Typical Values
1	Kinematic Viscosity mm ² / Sec At 40 °C At - 30 °C	ISO 3104 ISO 3104	≤ 12.0 ≤ 1800	10.50 1050
2	Flash Point , °C , PMCC	ISO 2719	≥ 140	148
3	Pour Point , °C ,	ISO 3016	≤ - 40	- 42
4	Appearance	IEC 60296	Clear, free from sediment & suspended matter	Clear, free from sediment & suspended matter
5	Density at 20 °C , g / ml Density at 29.5 °C , kg / dm ³	ISO 3675/ ISO 12185 ISO 3675/ ISO 12185	≤ 0.895 ≤ 0.890	0.842 0.838
6	Interfacial Tension at 25 °C , N / m	ISO 6295/ ASTM D971	≥ 0.04	0.047
7	Breakdown Voltage :			
a)	as delivered, KV	IEC 60156	≥ 30 kV(rms)	40 - 60
b)	after treatment, KV	IEC 60156	≥ 70 kV(rms)	> 72
8	Dissipation Factor at 90 °C (40 - 60 Hz)	IEC 60247	≤ 0.005	0.00015
9	Neutralisation Value , mg KOH / g	IEC 62021-1	≤ 0.01	0.0014
10	Corrosive Sulphur	DIN 51353	Non - Corrosive	Non corrosive
11	Water Content , mg / kg	IEC 60814	≤ 30 (in bulk supply) ≤ 40 (in drum supply)	14 12
12	Anti - oxidant additives	IEC 60666	Not Detectable	Not detectable
13	Oxidation stability (164 Hrs. at 120°C)	As per IEC 61125		
i)	Neutralisation Value , mg KOH / g	Method C	≤ 1.2	0.24
ii)	Sludge % by mass		≤ 0.8	0.05
14	PCA Content	BS 2000 Part 346	≤ 1 %	0.25
15	Impulse withstand Level, kVp.	IEC 60897	≥ 145	180
16	Gassing Tendency at 50Hz after 120 min, mm ³ /min	IEC 60628 (Method A)	≤ 5	+ 2.7


Authorised Signatory